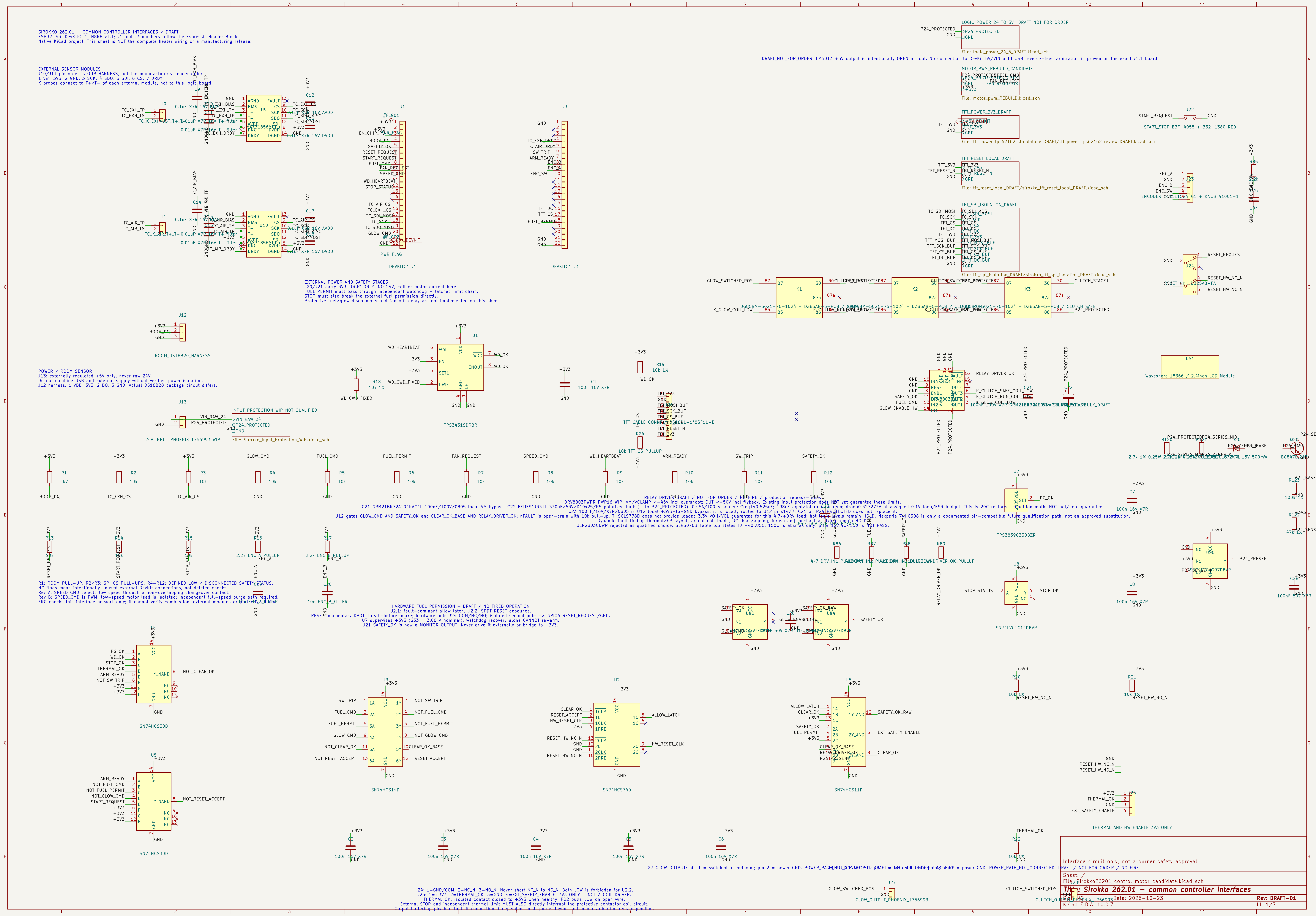

LOGIC_POWER_24_TO_5V__DRAFT_NOT_FOR_ORDER
File: logic_power_24_5_DRAFT.kicad_sch
MOTOR_PWM_REBUILD_CANDIDATE
File: motor_pwm_REBUILD.kicad_sch
TFT_POWER_3V3_DRAFT
File: tft_power_tps62162_standalone_DRAFT/tft_power_tps62162_review_DRAFT.kicad_sch
TFT_RESET_LOCAL_DRAFT
File: tft_reset_local_DRAFT/sirokko_tft_reset_local_DRAFT.kicad_sch
TFT_SPI_ISOLATION_DRAFT
File: tft_spi_isolation_DRAFT/sirokko_tft_spi_isolation_DRAFT.kicad_sch
INPUT_PROTECTION_WIP_NOT_QUALIFIED
File: Sirokko_Input_Protection_WIP.kicad_sch
J1 DEVKITC1_J1
J3 DEVKITC1_J3
J10
J11
U9 MAX31856MUD+
U10 MAX31856MUD+
J12 ROOM_DS18B20_HARNESS
J13 24V_INPUT_PHOENIX_1756993_WIP
U1 TPS3431SDRBR
K1
K2
K3
DG85BM-5021-76-1024 + DZ85AB-5-PCB / CLUTCH_RUN
DG85BM-5021-76-1024 + DZ85AB-5-PCB / CLUTCH_SAFE
U12
U7 TPS3839G33DBZR
U8 SN74LVC1G14DBVR
U20
U4 SN74HCS30D
U5 SN74HCS30D
U3 SN74HCS14D
U2 SN74HCS74D
U6 SN74HCS11D
J22 START_STOP B3F-4055 + B32-1380 RED
J23 ENCODER
J24 RESET
J25 THERMAL_AND_HW_ENABLE_3V3_ONLY
J27 GLOW_OUTPUT_PHOENIX_1756993
J28 CLUTCH_OUTPUT_PHOENIX_1756993
DS1 Waveshare 18366 / 2.4inch LCD Module
TFT CABLE CONN
PWR_FLAG
Q20 BC847B

SIROKKO 262.01 – COMMON CONTROLLER INTERFACES / DRAFT
ESP32-S3-DevKitC-1-N8R8 v1.1; J1 and J3 numbers follow the Espressif Header Block.
Native KiCad project. This sheet is NOT the complete heater wiring or a manufacturing release.

EXTERNAL SENSOR MODULES
J10/J11 pin order is OUR HARNESS, not the manufacturer's header order.
1 Vin=3V3; 2 GND; 3 SCK; 4 SDO; 5 SDI; 6 CS; 7 DRDY.
K probes connect to T+/T- of each external module, not to this logic board.

DRAFT_NOT_FOR_ORDER: LM5013 +5V output is intentionally OPEN at root. No connection to DevKit 5V/VIN until USB reverse-feed arbitration is proven on the exact v1.1 board.

EXTERNAL POWER AND SAFETY STAGES
J20/J21 carry 3V3 LOGIC ONLY. NO 24V, coil or motor current here.
FUEL_PERMIT must pass through independent watchdog + latched limit chain.
STOP must also break the external fuel permission directly.
Protective fuel/glow disconnects and fan off-delay are not implemented on this sheet.

POWER / ROOM SENSOR
J13: externally regulated +5V only, never raw 24V.
Do not combine USB and external supply without verified power isolation.
J12 harness: 1 VDD=3V3; 2 DQ; 3 GND. Actual DS18B20 package pinout differs.

RELAY DRIVER DRAFT / NOT FOR ORDER / NO FIRE / production_release=false.
DRV8803PWPR PWP16 WIP: VM/VCLAMP <=45V incl overshoot; OUT <=50V incl flyback. Existing input protection does NOT yet guarantee these limits.
C21 GRM21BR72A104KAC4L 100nF/100V/0805 local VM bypass. C22 EEUFS1J331L 330uF/63V/D10x25/P5 polarized bulk (+ to P24_PROTECTED). 0.45A/100us screen: Creq140.625uF; 198uF aged/tolerance screen; droop0.327273V at assigned 0.1V loop/ESR budget. This is 20C restored-condition math, NOT hot/cold guarantee.
C23 100nF/16V/X7R/0805 is U12 local +3V3-to-GND bypass; it is locally routed to U12 pins14/7. C21 on P24_PROTECTED does not replace it.
U12 gates GLOW_CMD AND SAFETY_OK and CLEAR_OK_BASE AND RELAY_DRIVER_OK; nFAULT is open-drain with 10k pull-up. TI SCLS778D does not provide loaded 3.3V VOH/VOL guarantee for this 4.7k+DRV load; hot logic levels remain HOLD. Nexperia 74HCS08 is only a documented pin-compatible future qualification path, not an approved substitution.
Dynamic fault timing, thermal/EP layout, actual coil loads, DC-bias/ageing, inrush and mechanical fixing remain HOLD.
ULN2803CDWR rejected as qualified choice: SLRS076B Table 5.3 states TJ -40..85C; 150C is absmax only; prior 127.4C<150 is NOT PASS.

R1: ROOM PULL-UP. R2/R3: SPI CS PULL-UPS. R4-R12: DEFINED LOW / DISCONNECTED SAFETY STATUS.
NC flags mean intentionally unused external DevKit connections, not deleted checks.
Rev A: SPEED_CMD selects low speed through a non-overlapping changeover contact.
Rev B: SPEED_CMD is PWM; low-speed motor lead is isolated; independent full-speed purge path required.
ERC checks this interface network only; it cannot verify combustion, external modules or protective timing.

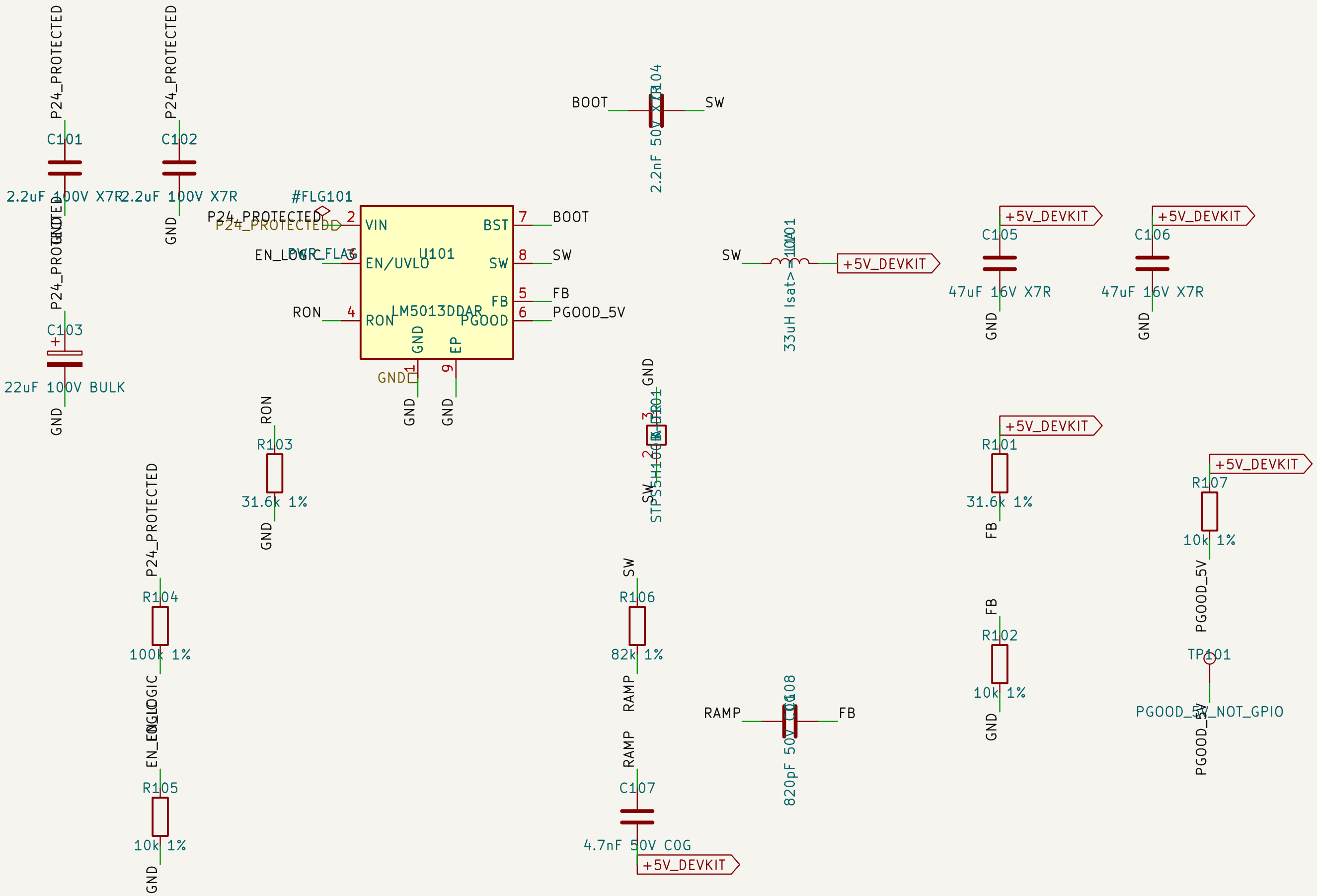
HARDWARE FUEL PERMISSION – DRAFT / NO FIRED OPERATION
U2.1: fault-dominant allow latch. U2.2: SPDT RESET debounce.
RESET: momentary DPDT, break-before-make; hardware pole J24 COM/NC/NO; isolated second pole -> GPIO6 RESET_REQUEST/GND.
U7 supervises +3V3 (G33 = 3.08 V nominal); watchdog recovery alone CANNOT re-arm.
J21 SAFETY_OK is now a MONITOR OUTPUT. Never drive it externally or bridge to +3V3.

J27 GLOW OUTPUT: pin 1 = switched + endpoint; pin 2 = power GND. POWER_PATH_NOT_CONNECTED. DRAFT / NOT FOR ORDER / NO FIRE.
J28 CLUTCH OUTPUT: pin 1 = switched + endpoint; pin 2 = power GND. POWER_PATH_NOT_CONNECTED. DRAFT / NOT FOR ORDER / NO FIRE.

J24: 1=GND/COM, 2=NC_N, 3=NO_N. Never short NC_N to NO_N. Both LOW is forbidden for U2.2.
J25: 1=+3V3, 2=THERMAL_OK, 3=GND, 4=EXT_SAFETY_ENABLE. 3V3 ONLY – NOT A COIL DRIVER.
THERMAL_OK: isolated contact closed to +3V3 when healthy; R22 pulls LOW on open wire.
External STOP and independent thermal limit MUST ALSO directly interrupt the protective contactor coil circuit.
Output buffering, physical fuel disconnection, independent post-purge, layout and bench validation remain pending.

Interface circuit only; not a burner safety approval

Sheet: /
File: Sirokko26201_control_motor_candidate.kicad_sch
Title: Sirokko 262.01 – common controller interfaces
Size: A2
Date: 2026-10-23
Rev: DRAFT-01
KiCad E.D.A. 10.0.7
Id: 1/7



тание логики 24 В → 5 В / 1 А. DRAFT, НЕ ДЛЯ ПРОИЗВОДСТВА.
5013, SNVSCB3 (10.2022), §6-9.2.2.7. Расчётный вход 18-30 В.
о ПОСЛЕ входной защиты; выбросы должны быть ограничены до ≤50 В.
к DevKit и логике на общей PCB. Это не внешний вход 5 В.
онтрольная точка, НЕ GPIO ESP32; не заменяет контроль наличия 24 В.
отключение обратной подпитки USB и силовые каналы здесь ещё не реализованы.
ые места требуют окончательной квалификации; особенно L101 и D101.
щё не выполнена; это отдельный проектный лист, не отдельная внешняя плата.

Sheet: /LOGIC_POWER_24_TO_5V__DRAFT_NOT_FOR_ORDER/
File: logic_power_24_5_DRAFT.kicad_sch

Title:

Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.7
Id: 2/7

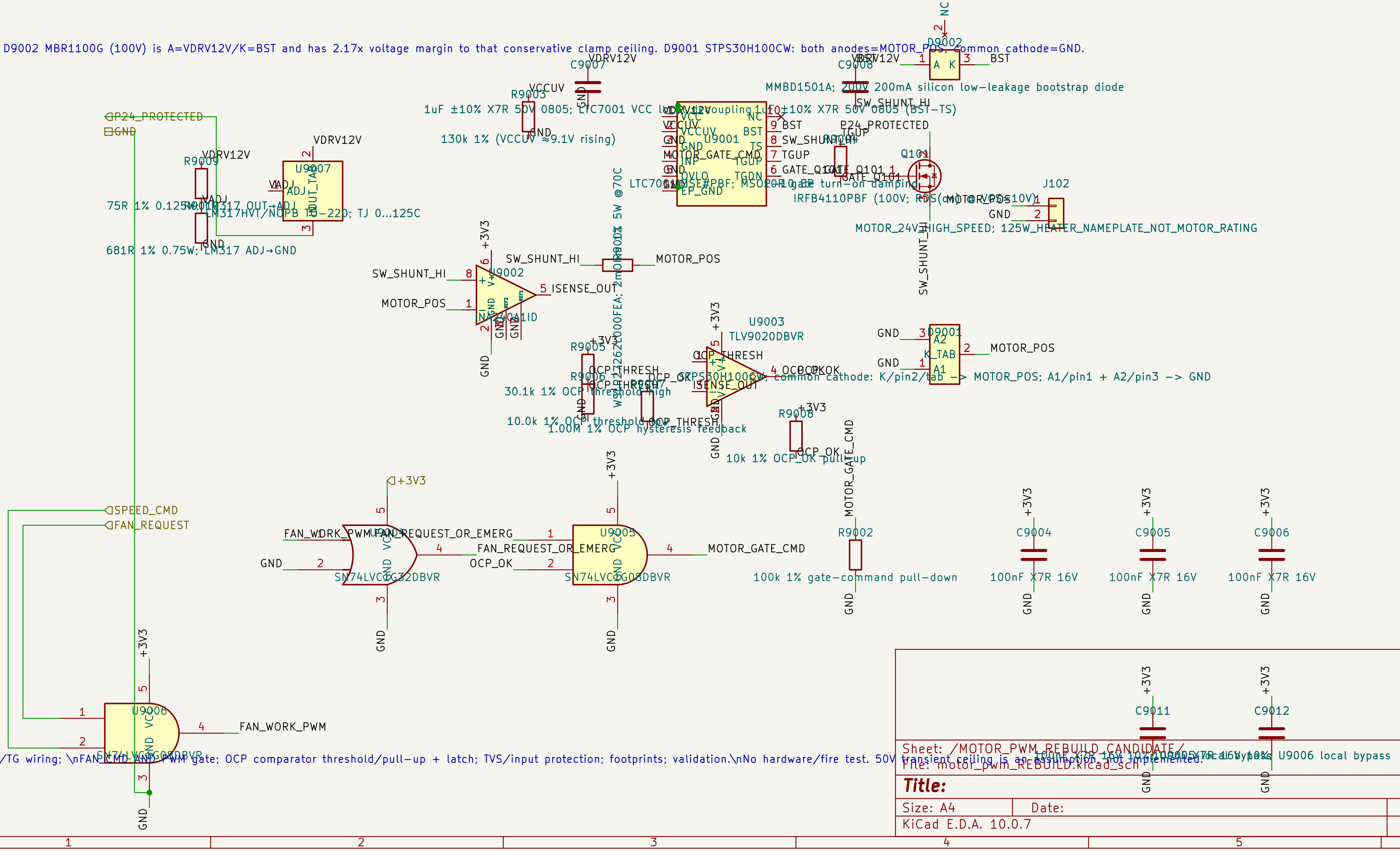
LOGIC CONTRACT – DRAFT, not bench qualified.
INP = ((FAN_CMD AND PWM_SPEED_CMD) OR FAN_EMERG_3V3) AND OCP_OK.
R9002=100k pulls INP LOW when +3V3/gates are off; VIN/VDRV may remain.
U9003 TLV9020DBVR: open–drain OUT4; IN+1, IN–3, V–2, V+5.
Overcurrent pulls OCP_OK LOW and inhibits both PWM and emergency command.
Verify physical shutdown timing and current decay on a safe bench.

CURRENT INTERFACE – DRAFT: 15 A continuous design limit, not measured motor current.
Logic design range +3V3=3.135...3.465 V; 3.0...3.6 V is an additional stress calculation.
Nominal ideal OCP at 3.3 V: trip 21.081 A, reset 20.471 A. These are NOT qualified limits.
See OCP_TLV9020_REVIEW_RU.md and ocp_threshold_model.json for assumptions.
Coordinate branch fuse, wiring, connector, MOSFET SOA and actual motor startup before release.

ed: A1(1)+A2(3)=GND; K/tab(2)=MOTOR_POS.
ualification stating reverse polarity is VALID.
ermal/SOA and input protection NOT qualified.

78125Hz; exact driver output–current, bootstrap recharge impedance, switching loss and thermal limits require bench capture before any dynamic–PWM maximum is released. Static OFF/ON is the only pre–bench permitted state.

lification remains OPEN. D9002 MBR1100G (100V) is A=VDRV12V/K=BST and has 2.17x voltage margin to that conservative clamp ceiling. D9001 STPS30H100CW: both anodes=MOTOR_POS, common cathode=GND.

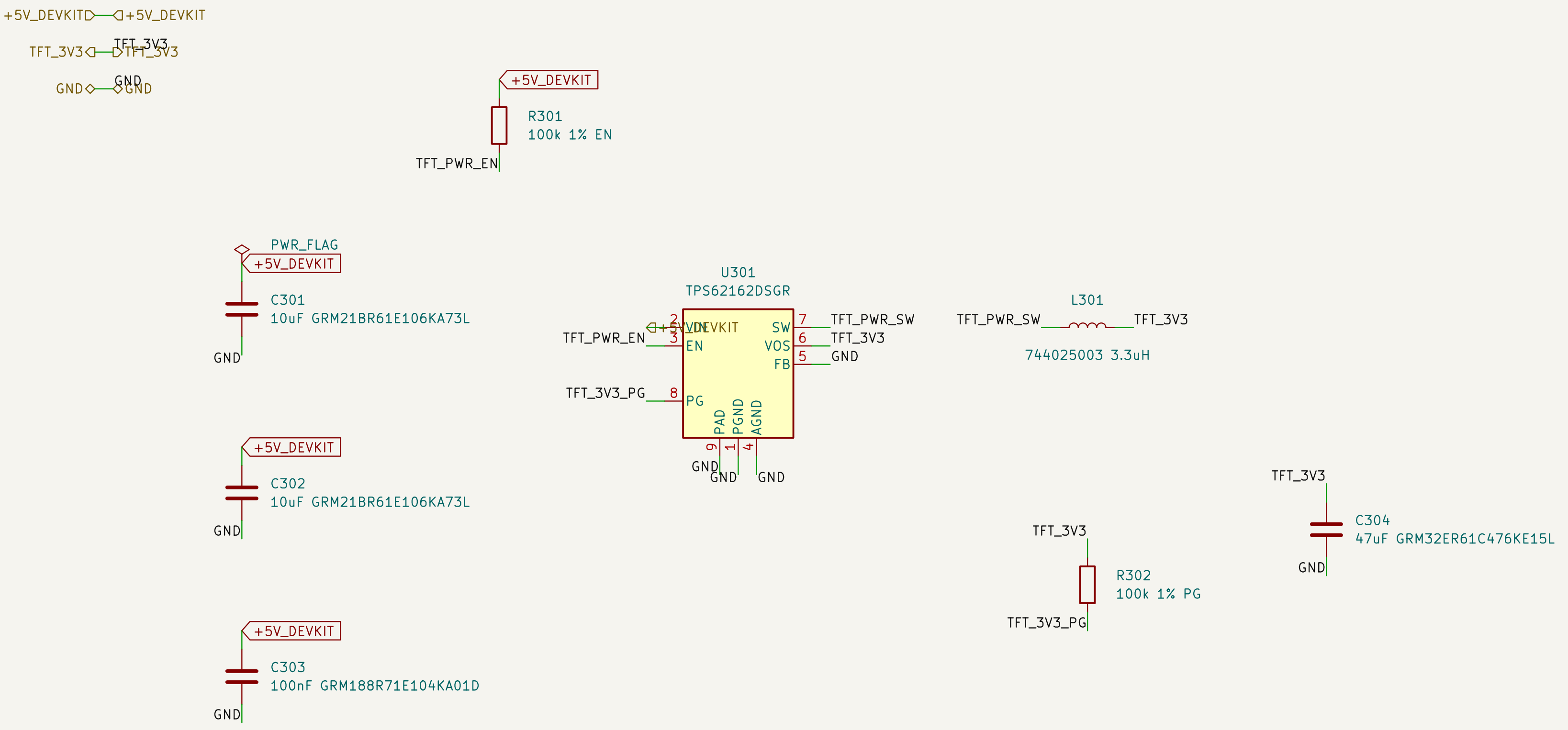

NC
D9002
VDRV12V
BST
MMBD1501A; 200V 200mA silicon low–leakage bootstrap diode
C9007
C9008
VCCUV
R9003
GND
1uF ±10% X7R 50V 0805; LTC7001 VCC local decoupling
1uF ±10% X7R 50V 0805 (BST–TS)
130k 1% (VCCUV ≈9.1V rising)
U9001
VCC
VCCUV
GND
INP
OVLO
EP_GND
NC
BST
TS
TGUP
TGDN
MOTOR_GATE_CMD
SW_SHUNT_HI
LTC7001MSE#PBF; MSOP–10 EP
R9004
10R gate turn–on damping
GATE_Q101
P24_PROTECTED
Q101
IRFB4110PBF (100V; RDS(on)@VGS=10V)
J102
MOTOR_POS
MOTOR_24V_HIGH_SPEED; 125W_HEATER_NAMEPLATE_NOT_MOTOR_RATING
P24_PROTECTED
GND
VDRV12V
U9007
ADJ
VOUT_TAB
R9009
R9010
VADJ
75R 1% 0.125W; LM317 OUT→ADJ
LM317HVT/NOPB TO–220; TJ 0...125C
681R 1% 0.75W; LM317 ADJ→GND
+3V3
U9002
ISENSE_OUT
INA240A1ID
R9001
WSL2512...; 2mOhm 1% 5W @70C
U9003
TLV9020DBVR
OCP_THRESH
OCP_OK
ISENSE_OUT
R9005
R9006
R9007
R9008
30.1k 1% OCP threshold high
10.0k 1% OCP threshold low
1.00M 1% OCP hysteresis feedback
10k 1% OCP_OK pull–up
D9001
A2
K_TAB
A1
STPS30H100CW; common cathode: K/pin2/tab –> MOTOR_POS; A1/pin1 + A2/pin3 –> GND
SPEED_CMD
FAN_REQUEST
FAN_WORK_PWM
FAN_REQUEST_OR_EMERG
U9004
SN74LVC1G32DBVR
U9005
SN74LVC1G08DBVR
MOTOR_GATE_CMD
R9002
100k 1% gate–command pull–down
C9004
C9005
C9006
100nF X7R 16V
U9006
SN74LVC1G08DBVR
C9011
C9012
100nF X7R 16V 10% 0805; local bypass
U9006 local bypass

rap; U9001 VCCUV/OVLO/TG wiring; \nFAN_CMD AND PWM gate; OCP comparator threshold/pull–up + latch; TVS/input protection; footprints; validation.\nNo hardware/fire test. 50V transient ceiling is an assumption, not implemented.

Sheet: /MOTOR_PWM_REBUILD_CANDIDATE/
File: motor_pwm_REBUILD.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 3/7

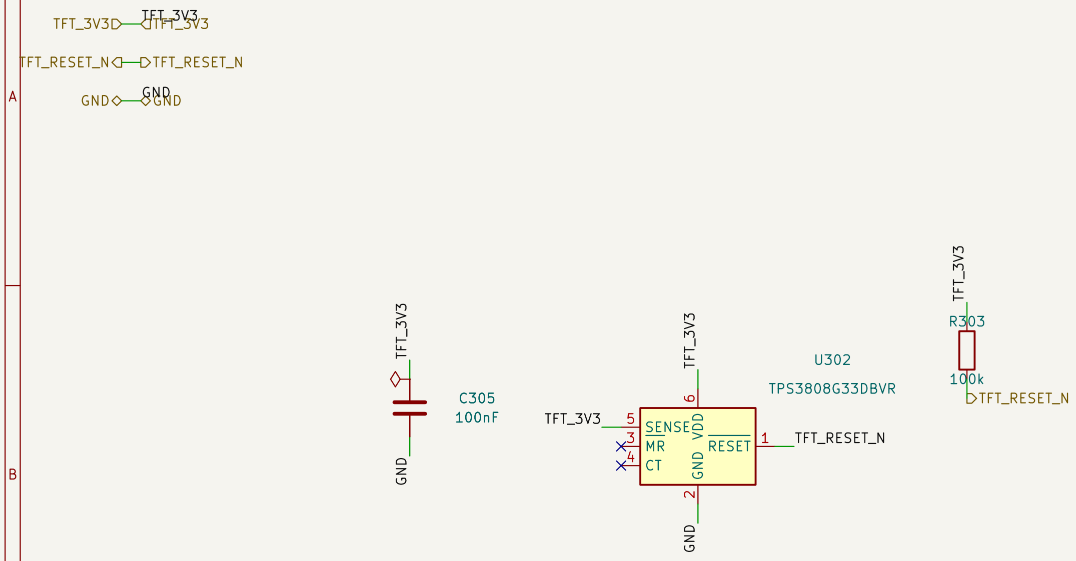


DRAFT / NOT FOR ORDER / NO FIRE
STANDALONE TFT POWER REVIEW – NOT CONNECTED TO ROOT PCB
External test supply: +5V_CARRIER 4.75..5.25V and GND; PWR_FLAG declares this external source ONLY.
TFT load 0..0.4A is a design assumption, not measured module current.
C301/C302: 10uF each,25V,X5R,0805; C304:47uF,16V,X5R,1210.
Ceff/DC bias,thermal,PG/EN ramp,SPI backfeed,layout and full5V budget NOT qualified.
PG is diagnostic open-drain, NOT an approved GPIO assignment or safety enable.
Kelvin VOS at C304+; VIN caps at VIN/PGND; AGND/PGND/EP single point.
No main SCH/PCB edits. General routing and firmware HOLD remain.
Do not import automatically: project ref collision / footprints must be reviewed.

Sheet: /TFT_POWER_3V3_DRAFT/
File: tft_power_tps62162_review_DRAFT.kicad_sch

Title:

Size: A4 Date: **Rev:**
KiCad E.D.A. 10.0.7 Id: 4/7

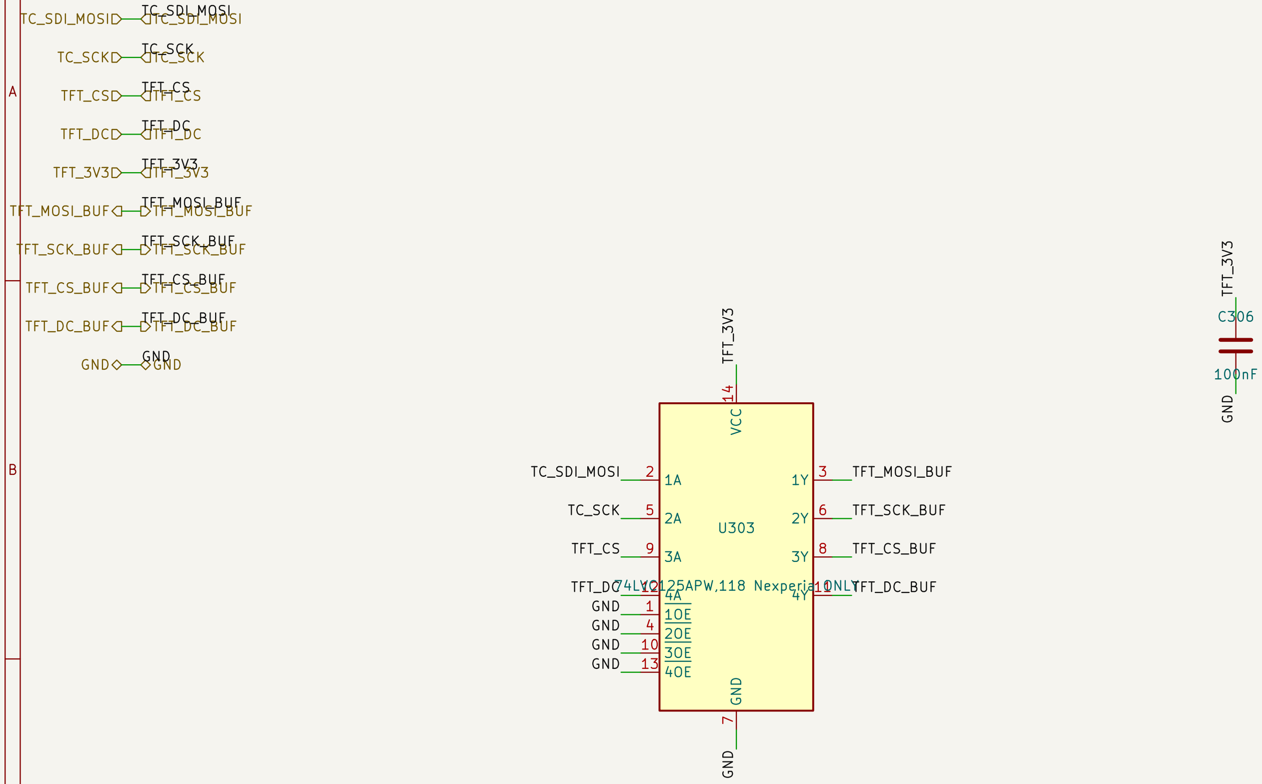


DRAFT / NOT FOR ORDER / NO FIRE
LOCAL TFT RESET REVIEW – NOT INTEGRATED
U302: TFT_3V3 supply + SENSE; RESET –> module RST only.
MR open: internal pull–up. CT open: 12...28 ms (20 ms typ).
R303 = RC0603FR–07100KL; C305 = GRM188R71E104KA01D.
PWR_FLAG declares EXTERNAL test supply / return, not board integration.
NO connection to ESP32 EN, safety RESET, or a GPIO.
Firmware reinitialization after TFT–only power loss remains OPEN.
TFT cable load / voltage thresholds / SPI backfeed remain unqualified.

Sheet: /TFT_RESET_LOCAL_DRAFT/
File: sirokko_tft_reset_local_DRAFT.kicad_sch

Title: TFT reset candidate – NOT INTEGRATED

Size: A4
Date:
Rev: DRAFT–01
KiCad E.D.A. 10.0.7
Id: 5/7

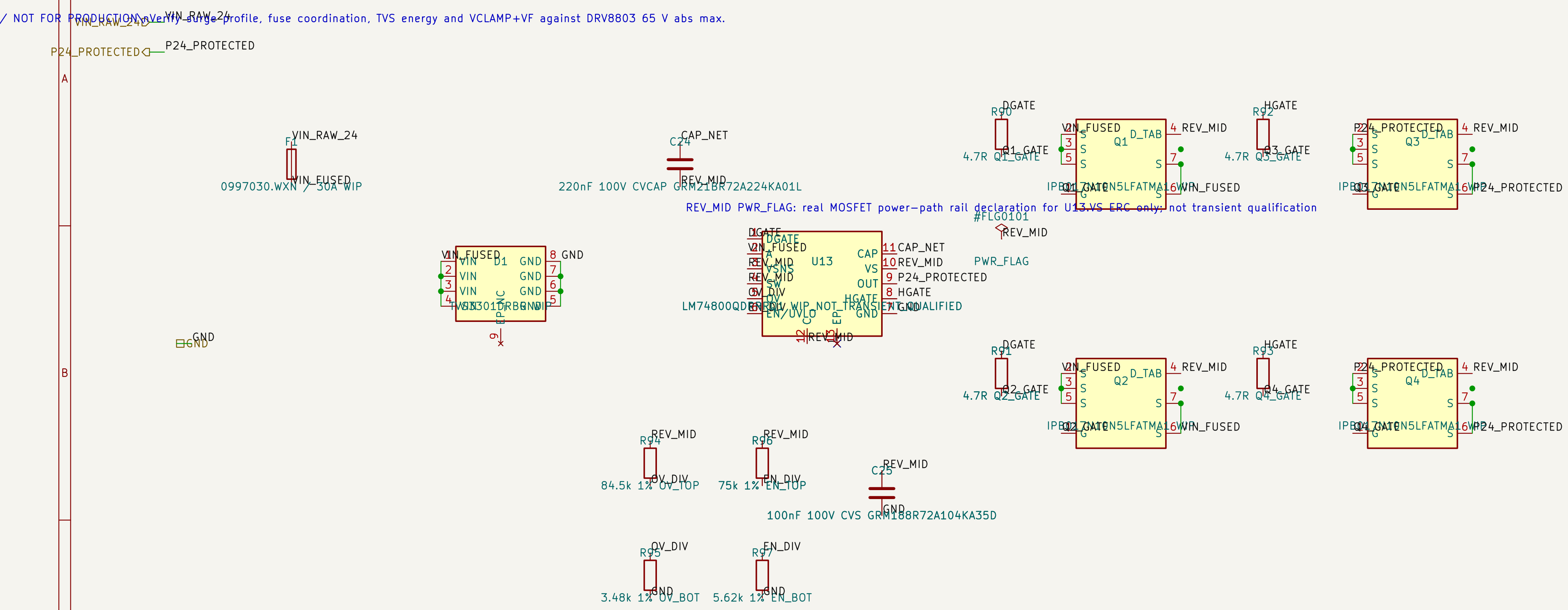


DRAFT / NOT INTEGRATED / NOT FOR ORDER / NO FIRE
TFT branch ONLY: power U303 from TFT_3V3, NOT MCU_3V3.
Nexperia 74LVC125APW,118 ONLY; do not substitute generic 125.
External supply / signal sources are outside this review sheet.
Required before integration: defined MCU-HiZ input levels,
source termination + cable SI, all power-ramp/dip cases,
TFT-local reset + firmware reinitialization.
RST and BL must stay in TFT power domain, not in this buffer.
No root PCB/SCH/BOM/firmware changes authorized by this sheet.

Sheet: /TFT_SPI_ISOLATION_DRAFT/
File: sirokko_tft_spi_isolation_DRAFT.kicad_sch

Title:

Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.7
Id: 6/7



/ NOT FOR PRODUCTION\nVerify surge profile, fuse coordination, TVS energy and VCLAMP+VF against DRV8803 65 V abs max.

REV_MID PWR_FLAG: real MOSFET power−path rail declaration for U13.VS ERC only; not transient qualification

Sheet: /INPUT_PROTECTION_WIP_NOT_QUALIFIED/
File: Sirokko_Input_Protection_WIP.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 7/7